

Evaluation of foundation model performance in chest radiology tasks based on pre-training data modality

Taejoon Park

21 August 2026

This report evaluates whether chest X-ray (CXR) specific foundation models outperform general biomedical models on thoracic radiology tasks. Using the MIMIC-CXR dataset, we compared six architectures—including the generalist MedSigLIP alongside CXR-specific models like CheXagent, CXR Foundation, and BioViL-T—across linear probing, zero-shot classification, and cross-modal retrieval. Our results demonstrate that the general biomedical model, MedSigLIP, achieved linear probing and zero-shot classification performance that was insignificantly divergent from its CXR-specific counterparts. Furthermore, MedSigLIP significantly outperformed domain-specific models in cross-modal retrieval tasks, suggesting that broad biomedical pre-training builds a more resilient semantic space that resists overfitting to formulaic radiology reports. Sensitivity analyses revealed that multi-layer perceptron classifiers did not consistently improve upon linear regression, confirming that these models natively produce cleanly organized, linearly separable latent spaces. Ultimately, this report concludes that maximizing pre-training data diversity yields robust foundational primitives that are highly competitive with—and in some tasks superior to—strictly domain-aligned representations.

Introduction

Chest X-ray (CXR) imaging remains a fundamentally accessible and ubiquitous diagnostic modality utilized across clinical environments of varying resource levels. Radiographically, this modality captures complex, low-level features that are indicative of a broad spectrum of thoracic pathologies. Because CXRs are processed in extraordinarily large volumes by practicing radiologists, artificial intelligence and deep learning methodologies have been actively pursued to mitigate this clinical burden.¹ These computational approaches not only seek to reduce the interpretive load on radiologists but also aim to enhance the recall rates of unexpected or subtle disease presentations.

A significant paradigm shift in this domain has been the advent of foundation models, which have emerged as a powerful advancement in medical image analysis. By leveraging massive, diverse pre-training datasets, these models develop robust latent representations that can be efficiently fine-tuned for a multitude of downstream clinical

tasks.ⁱⁱ However, as the spectrum of medical foundation models expands—ranging from general vision models to organ- and task-specific architectures—the nature of the pre-training data domain has become a primary distinguishing factor.

This evolution raises a critical, unresolved question in the field: do foundation models pre-trained exclusively on chest X-ray specific data inherently outperform models pre-trained on broader, general biomedical information? To address this, our study systematically evaluates a suite of state-of-the-art foundation models. We compare generalist models pre-trained on diverse biomedical text-image pairs, such as MedSigLIPⁱⁱⁱ, against domain-specific architectures including CheXagent^{iv}, CXR Foundation^v, CheXFound^{vi}, EVA-X^{vii}, and BioViL-T^{viii}.

By assessing model performance across structured label classification, zero-shot classification, and complex image-report retrieval tasks, this report investigates the performance boundaries dictated by pre-training data modalities. Ultimately, this research seeks to determine whether domain-specific alignment offers a definitive advantage over the diverse visual and semantic primitives acquired through broad biomedical pre-training.

Methods

Dataset

All evaluations were conducted on the MIMIC-CXR-JPG (v2.1.0)^{ix} and MIMIC-CXR (v2.1.0)^x datasets accessed via PhysioNet^{xi} following credentialled access approval. The dataset comprises 377,110 de-identified chest radiographs corresponding to 227,835 clinical studies collected from the Beth Israel Deaconess Medical Centre.^{xii}

Quality Control

To remove severe artefactual or corrupt images, an automated quality control pipeline was established:

1. Radiographs were initially ranked based on extreme image statistics, including the lowest half-image mean value, lowest pixel variance, and extreme mean pixel values.
2. A balanced sub-dataset (n = 50) composed of clean and problematic images was manually annotated to fine-tune an Ultralytics YOLO26-nano classification model.^{xiii}
3. Model rankings were combined with targeted manual verification to identify and soft-delete 85 severely corrupted images.

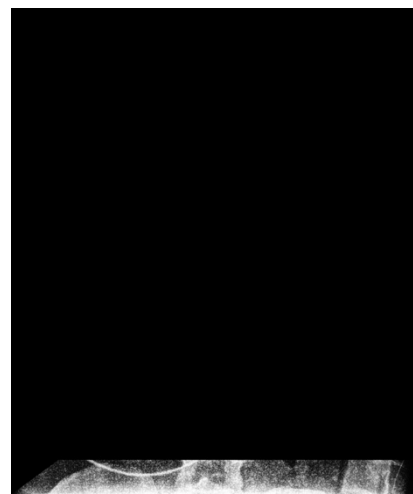


Figure 1, corrupted image likely, due to collimation error

Text and Label Processing

Radiology reports: Clinical text reports were parsed into distinct “Impression” and “Findings” sections.

Structured disease labels: Disease labels across 14 categories were extracted using CheXpert and NegBio natural language processing tools, encoded as positive (1), negative (0), uncertain (-1), or unmentioned (-1).^{xiv}

Cohort partitioning: Official patient-level test splits were strictly adhered to across all experiments to eliminate data contamination between training, validation, and testing phases.

Evaluated Foundation Model Architectures

Six state-of-the-art vision and vision-language foundation models were selected to contrast general biomedical pre-training against domain-specific chest radiology pre-training:

Model	Architecture	Pre-Training Modality & Domain Scope
MedSigLIP	SigLIP	CXR, Dermatology, Ophthalmology, Histopathology, CT/MRI slices, reports
CheXagent	SigLIP	CXR, reports
CXR Foundation	EfficientNet-L2 / BERT	CXR, reports
BioViL-T	ViT / ResNet-50 / BERT	CXR, reports
CheXFound	ViT-L	CXR
EVA-X	ViT-L	CXR

Table 1

All evaluations were performed with the highest performing variations of models as claimed by corresponding literature.

Computational Infrastructure and Inference

Inference and classifier head training were executed within isolated virtual environments tailored to each model’s specific dependency requirements. All computations were performed on the Tauon HPC server (MGH Partners network) equipped with four NVIDIA A6000 GPUs, accounting for approximately 50 total compute hours.

Image inputs were resized and normalised according to each model’s native preprocessing specifications. Text prompts and reports were tokenised using corresponding model tokenisers subject to respective sequence length limits. For vision-language models, feature vectors were extracted from the projected join image-text latent spaces.

Experimental Framework

Structured Disease Classification

To measure how effectively image embeddings separate disease states, a linear regression classifier head (PyTorch) was fitted on frozen image representations from each model to the 14 CheXpert disease labels.

Sensitivity analyses: To evaluate non-linear separability, Extreme Gradient Boosting (XGBoost) and Multi-Layer Perceptron (MLP) classification heads were benchmarked under various configurations (with and without L2 normalisation and Principal Component Analysis).

Hyperparameter search:

- Input dimensionality d_{in} ranged from 20 (PCA-compressed embeddings) to >5000 (early fusion representations).
- First hidden layer size: $k_1 = \alpha \cdot d_{in}$, where $\alpha \in \{0.5, 0.67, 1.0, 1.5, 2.0\}$
- Optional second hidden layer size: $k_2 = \max\left(\left\lfloor \frac{k_1}{2} \right\rfloor, 14\right)$ to avoid bottleneck compression below the 14 target classes.
- Dropout regularisation: $p \in \{0.2, 0.5\}$
- Optimisation: early stopping triggered after 5 consecutive epochs without a 0.0001 validation loss improvement

Evaluation metrics: Area under the receiver operating character curve (AUROC) and area under the precision-recall curve (AUPRC).

Zero-Shot Disease Classification

Zero-shot diagnostic performance was evaluated for image-text contrastive models (MedSigLIP, CXR Foundation, BioViL-T) by computing the cosine similarity between frozen image embeddings and custom text positive-negative prompt pairs in the joint latent space. Text prompt pairs were optimised to be similar in length and vocabulary to minimise semantic effects.

Disease	Positive prompt	Negative prompt
Atelectasis	“radiographic findings consistent with atelectasis”	“no evidence of atelectasis, lungs are fully inflated”
Cardiomegaly	“radiographic findings consistent with cardiomegaly, enlarged hearth silhouette”	“normal cardiac silhouette size, normal heart size”
Consolidation	“radiographic findings consistent with lung consolidation”	“no evidence of consolidation, lungs are clear”
Edema	“radiographic findings consistent with pulmonary edema, vascular congestion”	“no evidence of pulmonary edema, lung fields are clear”

Enlarged cardiomeastinum	“radiographic findings consistent with an enlarged cardiomeastinum”	“normal cardiomeastinal contour, no mediastinal widening”
Fracture	“radiographic findings consistent with a fracture”	“no evidence of acute fracture, intact bony structures”
Lung lesion	“radiographic findings consistent with a lung lesion, nodule, or mass”	“no evidence of pulmonary nodules or lung masses”
Lung opacity	“radiographic findings consistent with lung opacity”	“no evidence of abnormal lung opacities, lung fields are lucent”
Pleural effusion	“radiographic findings consistent with pleural effusion”	“sharp costophrenic angles, no evidence of pleural effusion”
Pleural other	“radiographic findings consistent with pleural thickening or fibrocalcific scarring”	“normal pleural space, no pleural thickening”
Pneumonia	“radiographic findings consistent with pneumonia, infectious infiltrate”	“no evidence of focal pneumonia or infectious infiltrate”
Pneumothorax	“radiographic findings consistent with pneumothorax, pleural air”	“no evidence of a pneumothorax, lung apex is intact”
Support devices	“presence of support devices, tubes, lines, or hardware”	“no support devices, medical lines, or tubes present”
No finding	“normal chest x-ray, clear lung fields, no acute cardiopulmonary abnormalities”	“abnormal radiographic findings, presence of acute cardiopulmonary disease”

Table 2

Evaluation metrics: Area under the receiver operating character curve (AUROC) and area under the precision-recall curve (AUPRC).

Cross-Modal Image-Report Retrieval

To evaluate cross-modal alignment, models were tasked with matching patient radiographs to their corresponding radiology reports and vice versa:

Multi-image/report handling: For clinical studies containing multiple views, the image representation was defined as the centroid of all image embeddings within the study. The report representation was calculated as the centroid of the embeddings for the “Impression” and “Findings” sections.

Evaluation metrics: Recall at k (R@1, R@5, R@10) and mean reciprocal rank (MRR).

Results

Dataset Demographics

Analysis of the MIMIC-CXR-JPG cohort confirmed unbalanced data distributions as expected with the respective natural rates of diseases, but suitable for robust foundation model evaluation. Diseases such as pleural other and fracture were far underrepresented yet present. Individual patients were not overrepresented to a problematic degree, and patient records were strictly partitioned across train, validation, and test splits to guarantee no data leakage. Furthermore, the burden of comorbidity within individual radiographs was distributed normally and did not present at extreme degrees that would confound model inference.

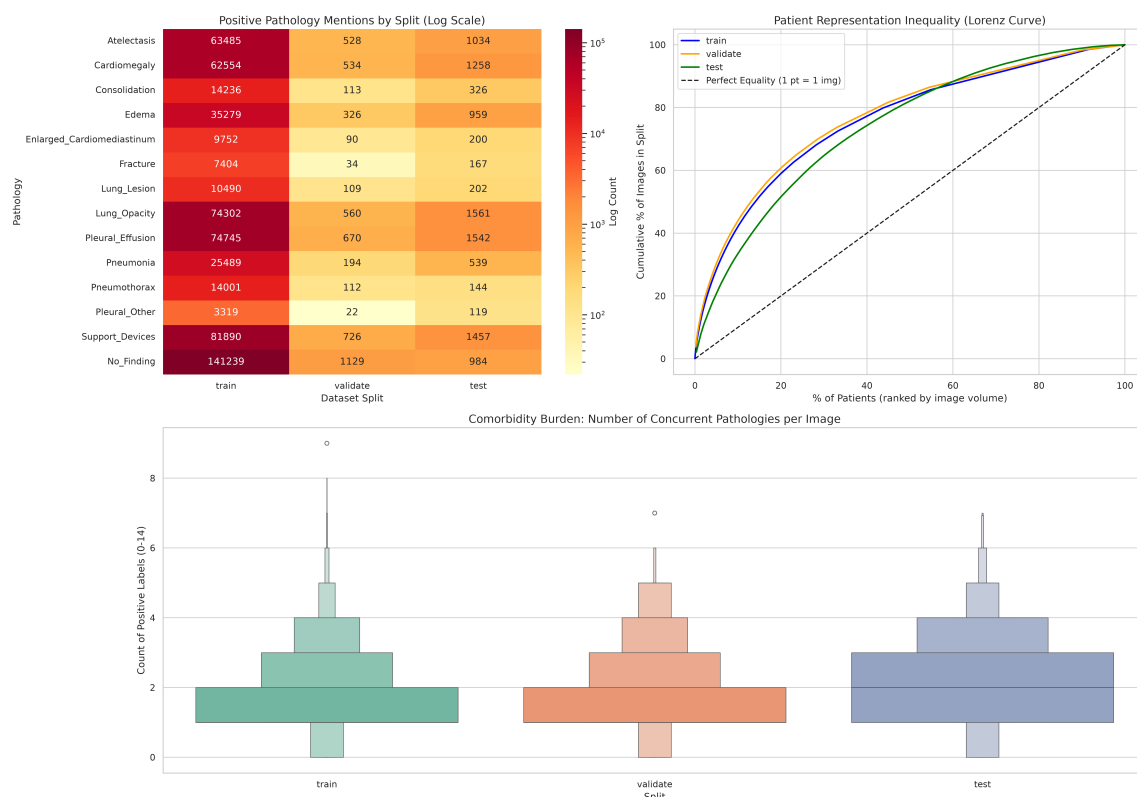


Figure 2. Top-left, positive disease mentions by dataset split. Data for certain diseases were sparse. Top-right, Lorenz curve of patient representation inequality in images. Approximately 50-60% of images were represented by 20% of patients. Bottom, comorbidity by dataset split.

Linear Probing

Linear probing on frozen image embeddings revealed that pre-training data domain does not strictly dictate classification performance. The general biomedical model, MedSigLIP, demonstrated similar performance in AUROC and AUPRC when compared to models pre-trained exclusively on chest X-ray domains. Notably, the domain-specific CheXagent architecture achieved marginally superior performance overall in linear probing evaluations.

Zero-Shot Classification and Cross-Modal Retrieval

Zero-shot disease classification performance remained highly comparable between the general biomedical domain and CXR-specific domains. However, in cross-modal retrieval tasks, the generalist MedSigLIP significantly outperformed domain-specific architectures.

In tasks matching corresponding patient images to their clinical reports (and vice versa), MedSigLIP achieved consistently higher retrieval metrics across all recall thresholds compared to CXR Foundation and BioViL-T.

Model	Pre-training domain	Macro AUROC	Macro AUPRC
MedSigLIP	General	0.7975121	0.3310816
CheXagent	CXR	0.82133162	0.38352254
CXR_Foundation	CXR	0.78225978	0.33365863
BioViL-T	CXR	0.63325776	0.19098472
CheXFound	CXR	0.79155024	0.33838169
EVA-X	CXR	0.78426876	0.31882967

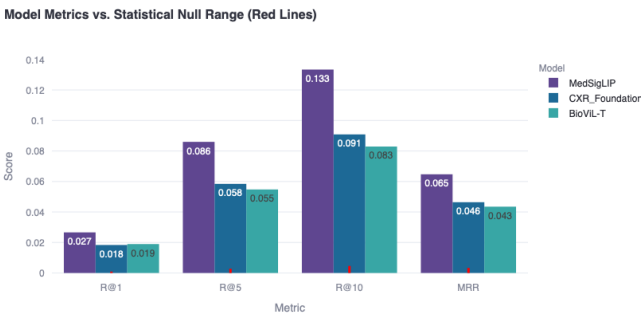


Figure 3. Left, Macro AUROC and AUPRC in linear probing. Right, recall metrics in image-to-report matching.

Secondary Outcomes

Latent space separability: During sensitivity analyses, non-linear Multi-Layer Perceptron (MLP) and Extreme Gradient Boosting (XGB) classification heads did not demonstrate consistently higher performance than simple linear regression across the embedded latent spaces.

Shared architectural weaknesses: Irrespective of the pre-training data modality or model scale, all evaluated foundation models exhibited shared vulnerabilities on specific disease categories. Most prominently, performance on atelectasis—despite being a well-represented disease with large visual features—was consistently poor across all models and classification methods.

Discussion

The comparative analysis of these foundation models yields critical insights into the role of pre-training data domains. Our primary finding is that MedSigLIP, a general biomedical model, demonstrated insignificant divergence in linear probing metrics compared to chest X-ray (CXR) specific architectures. This suggests that data diversity during pre-training is just as critical as data scale. By pre-training across diverse imaging modalities, models like MedSigLIP learn fundamental visual primitives that effectively transfer to thoracic

radiology. While CheXagent achieved marginally superior linear probing results—indicating that domain-specific alignment provides a slight advantage—the gap is remarkably narrow.

Furthermore, the sensitivity analyses confirmed our hypothesis regarding the embedded latent feature spaces. Non-linear multi-layer perceptron (MLP) and extreme gradient boosting (XGBoost) classifier heads failed to consistently outperform simple linear regression. This indicates that foundation model pre-training inherently yields cleanly organized, linearly separable latent spaces, rendering complex non-linear transformations unnecessary for downstream classification.

In cross-modal tasks, MedSigLIP significantly outperformed CXR-specific models in image-to-report retrieval. This disparity likely stems from CXR-only language models overfitting to the highly formulaic and templated nature of standard radiology reports. Conversely, pre-training on broader medical literature appears to create a richer, more flexible semantic space that resists such overfitting.

Lastly, all evaluated models shared poor performance on specific labels. Atelectasis (characterized by large features) and fractures (characterized by pixel-level features) both suffered. This points to a shared architectural weakness: the down-sampling of images may lose critical boundaries, while patch-wise tokenization may obscure large, continuous features.

Future Directions

To address these limitations and expand upon our findings, future research should focus on the following areas:

Dense Latent Probing: Evaluating token-level dense feature maps rather than globally pooled embeddings could improve spatial resolution, preserving small features and enhancing clinical explainability through phrase grounding.

Targeted Fine-Tuning: Utilizing efficient techniques like LoRA on curated datasets can improve performance on difficult disease labels (e.g., fractures) while preventing catastrophic forgetting by freezing base weights.

Hybrid Pre-Training: Implementing a two-stage approach—initial general biomedical pre-training followed by domain-specific pre-training using prototype-driven data selection^{xv}—could maximize data diversity, prevent latent space collapse, and avoid over-indexing on repetitive boilerplate reports.

Generative Tasks: Investigating whether CXR-specific foundation models outperform generalist models in structured report generation remains a critical next step.

External Validation: Future studies must evaluate whether generalist models demonstrate superior zero-shot and linear probing generalization when exposed to out-of-distribution institutional data.

All models and data used in this project are publicly available online at their respective hosting locations.

All code used in this project is available at

https://github.com/tpark747949/cxr_foundation_eval with documentation.

I would like to note my sincere gratitude to Professor Bongjin Lee for the project direction and guidance, and to Professor Jonghye Woo for the support and opportunity.

ⁱ Milam ME, Koo CW. The current status and future of FDA-approved artificial intelligence tools in chest radiology in the United States. *Clin Radiol*. 2023; 78(2):115-122. <https://doi.org/10.1016/j.crad.2022.08.135>.

ⁱⁱ Zhang S, Metaxas D. On the challenges and perspectives of foundation models for medical image analysis. *Med Image Anal*. 2024; 91:102996. <https://doi.org/10.1016/j.media.2023.102996>.

ⁱⁱⁱ Google. MedSigLIP. Health AI Developer Foundations. July 9, 2025. Accessed August 21, 2026.

<https://developers.google.com/health-ai-developer-foundations/medsiglip>.

^{iv} Chen Z, Varma M, Xu J, et al. A vision-language foundation model to enhance efficiency of chest X-ray interpretation. *arXiv preprint arXiv:2401.12208*. 2024. <https://doi.org/10.48550/arXiv.2401.12208>.

^v Google. CXR Foundation Model. Health AI Developer Foundations. June 5, 2026. Accessed August 21, 2026.

<https://developers.google.com/health-ai-developer-foundations/cxr-foundation>.

^{vi} Yang Z, Xu X, Zhang J, Wang G, Kalra MK, Yan P. Chest x-ray foundation model with global and local representations integration. *IEEE transactions on medical imaging*, 2025;44(12): 4787-4799.

<https://doi.org/10.48550/arXiv.2502.05142>.

^{vii} Yao J, Wang X, Song Y, et al. EVA-X: a foundation model for general chest x-ray analysis with self-supervised learning. *npj Digit. Med*. 2025;8: 678. <https://doi.org/10.1038/s41746-025-02032-z>.

^{viii} Bannur S, Hyland S, Liu Q, et al. (2023, June). Learning to exploit temporal structure for biomedical vision-language processing. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023:15016–15027. <https://doi.org/10.48550/arXiv.2301.04558>.

^{ix} Johnson AE, Lungren M, Peng Y, et al. MIMIC-CXR-JPG - chest radiographs with structured labels (version 2.1.0). *PhysioNet*. 2024; RRID:SCR_007345. <https://doi.org/10.13026/jsn5-t979>.

^x Johnson AE, Pollard T, Mark R, Berkowitz S., Horng S. MIMIC-CXR Database (version 2.1.0). *PhysioNet*. 2024; RRID:SCR_007345. <https://doi.org/10.13026/4jqj-jw95>.

^{xi} Johnson AE, Lungren M, Peng Y, et al. MIMIC-CXR-JPG - chest radiographs with structured labels (version 2.1.0). *PhysioNet*. 2024; RRID:SCR_007345. <https://doi.org/10.13026/jsn5-t979>.

^{xii} Johnson AE, Pollard TJ, Berkowitz S, Greenbaum NR, Lungren MP, Deng CY, Mark RG, Horng S. MIMIC-CXR: A large publicly available database of labeled chest radiographs. *arXiv preprint*. 2019. <https://doi.org/10.48550/arXiv.1901.07042>.

^{xiii} Ultralytics. Ultralytics YOLO26. Docs. Accessed August 21, 2026.

<https://docs.ultralytics.com/models/yolo26>.

^{xiv} Johnson AE, Lungren M, Peng Y, et al. MIMIC-CXR-JPG - chest radiographs with structured labels (version 2.1.0). *PhysioNet*. 2024; RRID:SCR_007345. <https://doi.org/10.13026/jsn5-t979>.

^{xv} Wang C, Zhang Y, Gao Y, et al. A data- and compute-efficient chest X-ray foundation model beyond aggressive scaling. *arXiv preprint*. 2026; arXiv:2602.22843 <https://doi.org/10.48550/arXiv.2602.22843>.